

# Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



TEAM NAME

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PROBLEM STATEMENT

**Problem Statement 1**

**AI-Powered Decision Intelligence Platform**

## Brief About the Idea: Annapurna AI

### The Cold-Chain Problem

#### ₹1.52 Lakh Crore Lost

Every year, India loses massive perishable produce in transit—enough to feed **200 million people**.

#### The 2 AM Nightmare

When a truck compressor fails on a highway, zero automation means a manual chain of calls. By the time anyone acts, **5,000 kg of mangoes (₹3,00,000)** turn to compost.

### The AI-Powered Solution

#### Annapurna Platform

An autonomous, self-healing logistics brain that transforms India's broken cold-chain without a single human in the loop.

- **Predicts Spoilage:** 14 days ahead via real-time IoT telemetry.
- **Auto-Rerouting:** Instantly redirects trucks upon failure detection.
- **AI Negotiation:** Runs AI-vs-AI price bidding to close emergency sales.

### Tech & Real Impact

#### 90-Second Recovery

Converts a **₹3,00,000 total loss** into a **₹2,55,000 recovery** in under 90 seconds.

#### POWERED BY GOOGLE CLOUD

Vertex AI Gemini 2.5 Flash • Google ADK Agents • Cloud Firestore • Firestore Vector Search • Vertex AI Text Embeddings • BigQuery ML • Cloud Run • Cloud Translate • Generative AI SDK

## How We Built It: Google Cloud Architecture

The production tech stack powering Annapurna's autonomous cold-chain rescue engine

### Vertex AI & ADK

#### Gemini 2.5 Flash

The central AI brain powering 8 capabilities including autonomous cargo routing, spoilage vision QC, and multi-agent negotiation.

#### Agent Dev Kit (ADK)

Powers the fleet decision agent using executable function tools to autonomously rewrite Firestore destinations.

### Cloud Firestore

#### Real-Time State

Syncs fleet telemetry and bids instantly across dashboards via onSnapshot listeners with atomic batch writes.

#### Native Vector Search

Powers Legal RAG. Embeds food regulations with text-embedding-004 and queries using cosine distance.

### BigQuery & ML

#### Telemetry Warehouse

Streams massive IoT sensor data with built-in natural-language-to-SQL analytics for real-time querying.

#### ARIMA\_PLUS Model

Predicts shipment spoilage risk 14 days ahead, training and serving ML directly inside the data warehouse.

### Cloud Run & More

#### Serverless Compute

Hosts the entire Next.js 15 application as a containerized, auto-scaling service (2 vCPU, 2GB RAM).

#### Cloud Translate API

Enables multi-language localization (Hindi, Thai, Vietnamese, etc.) for seamless JAPAC-wide deployments.

### Core Workflow: Detect → Reason → Decide → Act → Negotiate → Recover → Learn

IoT telemetry stream → anomaly detection → ADK decision agent routes around failure → pricing agent negotiates 3-round bid → atomic transaction completes.

**The Impact: Replaces 45-minute manual bottlenecks with 90-second AI decisions, recovering 70-85% of cargo value.**

## Opportunities

What makes Annapurna fundamentally different?

### Traditional Platforms Wait

Shows data on screens and waits for human decisions (FourKites alerts, Carrier GPS pins). Then YOU must figure out what to do.

### Annapurna Acts Autonomously

Annapurna doesn't wait. It executes real-time database writes, resolving cold-chain pipeline failures instantly without humans in the loop.

#### 1. Autonomous Action

Calls `reroute_truck()` and `alert_wholesaler()`. The AI doesn't just recommend—it executes real Firestore writes immediately.

#### 2. AI Negotiation

Two Gemini-powered agents negotiate penalty pricing in real-time. No brokers, no phone calls, no delays. Reaches fair market price in under 10 seconds.

#### 3. Unified GCP Stack

9 Cloud Services deeply integrated into one system. Vertex AI, ADK, Firestore, BigQuery, Cloud Run, Translate, and GenAI SDK power robust production.

#### 4. Native Vector RAG

Legal compliance reports generated via text-embedding-004. Uses Firestore native **findNearest (COSINE)** and Gemini. No external databases.

#### 5. 85% Recovery

Complete rescue pipeline from cold-chain detection to payment. Converts 100% loss into **85% recovery** in one autonomous flow taking under 90 seconds.

## List of Features Offered by the Solution

Comprehensive capabilities powering the autonomous cold-chain management platform

### Fleet Command Center

- Live GPS truck tracking & real-time IoT telemetry
- Autonomous AI routing via Gemini evaluations
- Nerve Center Terminal multi-agent decision logs
- Vision AI QC & Document AI data extraction
- AI fleet metrics & Carbon credit tokenization

### Wholesaler Marketplace

- Real-time emergency cargo feed via Firestore
- Full or partial quantity bidding system
- Live order lifecycle with automated transition sync
- Integrated Vision QC & Document AI portals
- Historical transaction & ledger management

### Autonomous AI Agents

- ADK fleet decision agents running via InMemoryRunner
- Automated rerouting & wholesaler alert dispatch
- 3-round adversarial NegotiationAgent exchange
- Continuous scanning, broadcasting & notification
- Gemini-generated real-time multi-agent logs

### Analytics & Intelligence

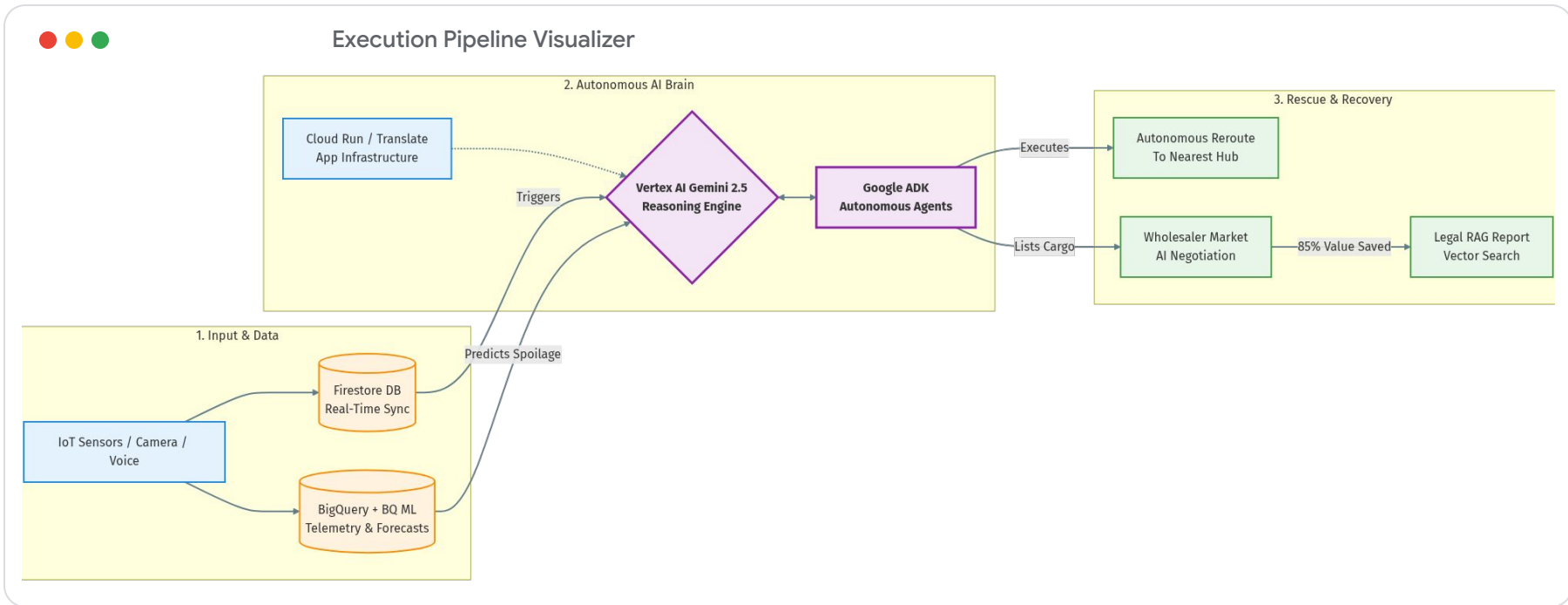
- Text-to-SQL: English questions to BigQuery SQL execution
- ARIMA\_PLUS: 14-day spoilage risk forecasting model
- Legal RAG: text-embedding-004 + Vector Search for compliance
- Driver voice command extraction for intents & entities

### Platform & Deployment

- Role-based authentication & secure geolocation
- JAPAC multi-language via Cloud Translate API
- Serverless Cloud Run hosting with auto-scaling
- Premium glassmorphic user interface & animations

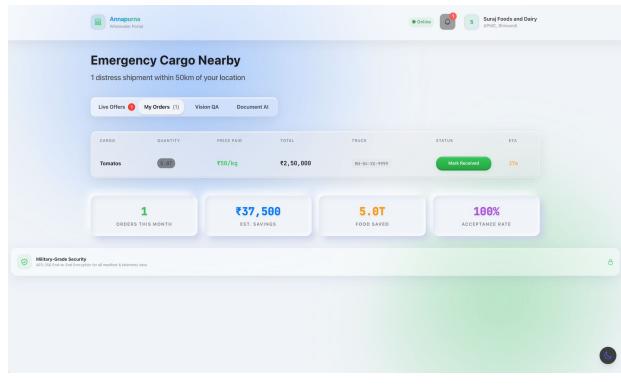
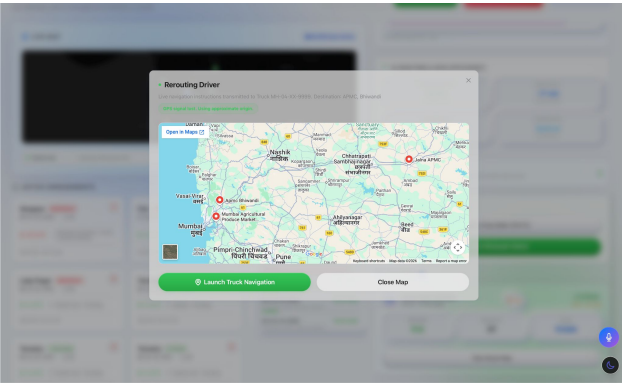
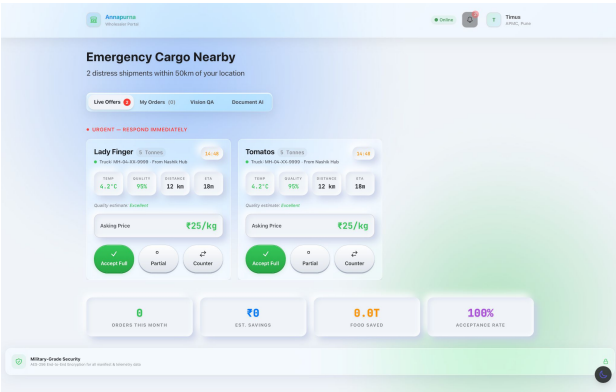
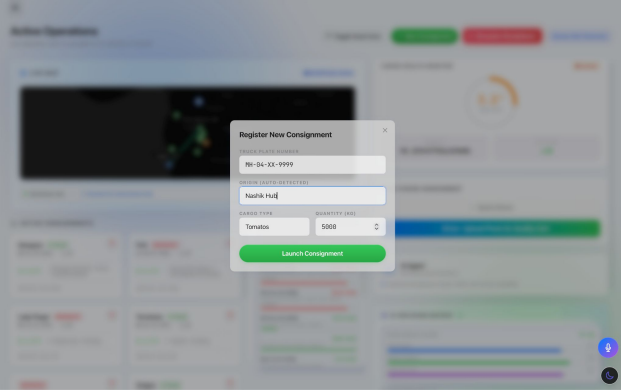
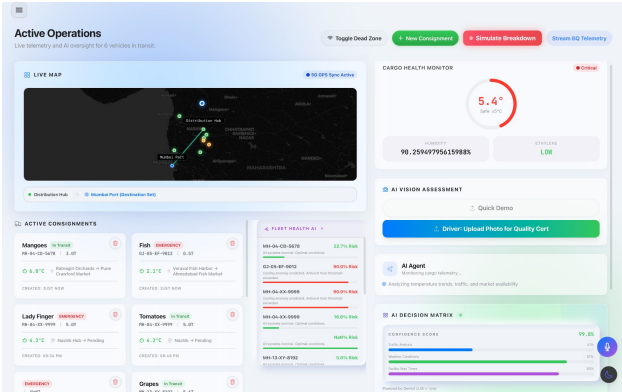
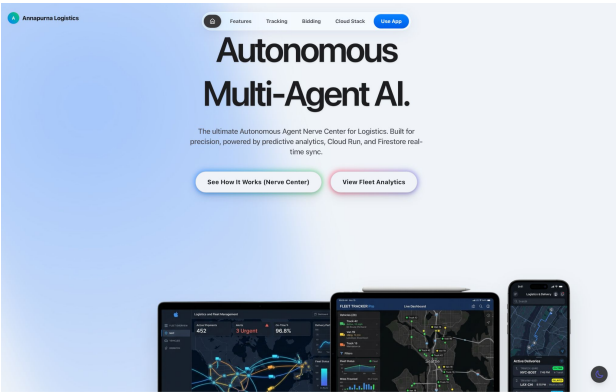
## Process Flow Diagram

End-to-end data pipelines and autonomous decision loops



# Wireframes & Mockups

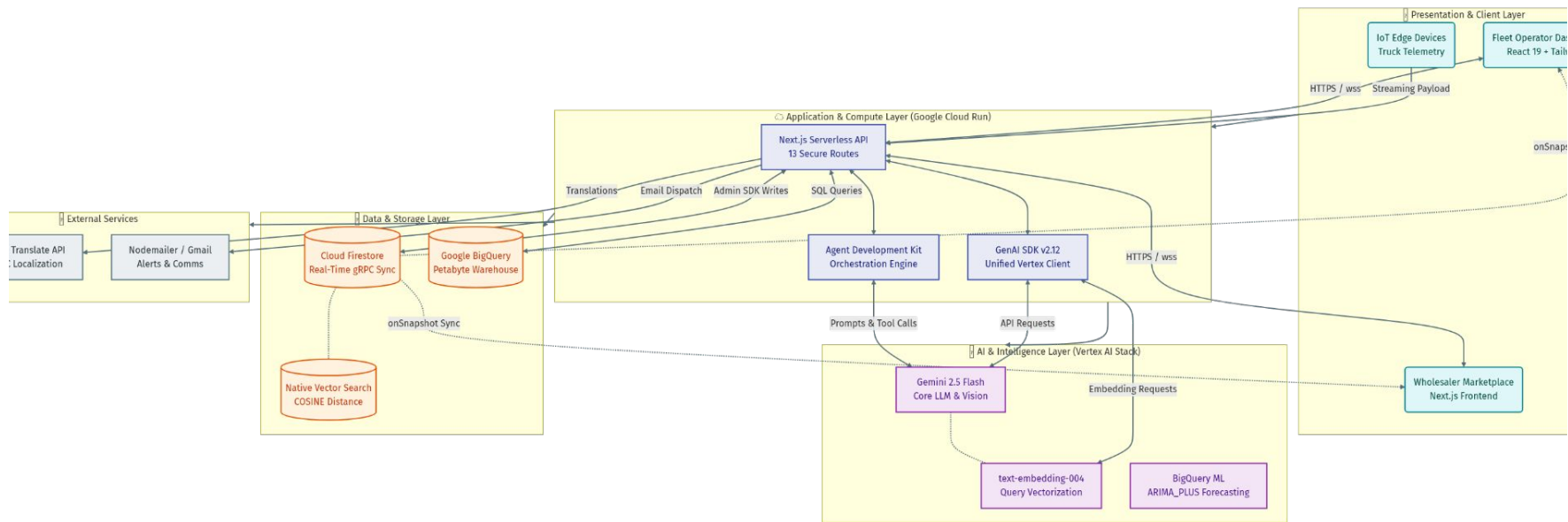
Visual interface walkthrough of the autonomous cold-chain management solution



# Architecture Diagram

System design and technology stack layout for the proposed solution

## Proposed Solution Architecture





## Technologies / Google Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

### AI & Agent Framework

#### 1. Vertex AI (Gemini 2.5 Flash)

The AI brain powering 8 capabilities (routing, negotiation, voice). Sub-2s responses.

#### 2. Google ADK (@google/adk v1.4)

Autonomous agent framework with real FunctionTool bindings executing backend functions.

#### 3. Generative AI SDK (v2.12)

Unified client with smart proxy. Single entry point for all Gemini capabilities.

### Real-Time Data & Search

#### 4. Cloud Firestore

Real-time DB with onSnapshot sync. Sub-100ms writes for instant cross-city dispatch updates.

#### 5. Native Vector Search

findNearest() cosine distance for legal RAG. Native execution where data lives.

#### 6. Vertex AI Embeddings

text-embedding-004 model generates vector queries perfectly optimized for Firestore.

### Scale, Analytics & Infra

#### 7. BigQuery + BQ ML

Telemetry warehouse with ARIMA\_PLUS forecasting for 14-day spoilage prediction.

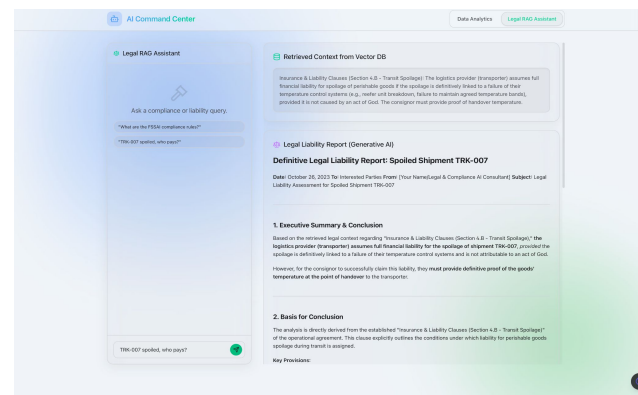
#### 8. Cloud Run

Auto-scaling Next.js containers. Scales 0 to 1000+ instances. Push to live in 60s.

#### 9. Cloud Translate API

Instant JAPAC multi-language support (Hindi, Japanese, Thai, Bahasa, Vietnamese).

**Why this stack scales:** Firestore auto-scales horizontally. Cloud Run handles burst traffic instantly. BigQuery manages petabytes without ops overhead, and stateless ADK agents process 10,000+ parallel trucks efficiently.



## Improvements done during prototype refinement phase

### System & UX Evolution

Data Layer: Hardcoded mock data → Live Firestore with real-time sync

Stakeholders: Single Fleet dashboard →

Multi-stakeholder platform

Deployment: localhost only → Cloud Run production scale

UX: Basic HTML → High Tier glassmorphism & maps

### AI & Analytics Engine

AI Engine: If/else check → ADK LlmAgent with Gemini 2.5

Agent System: Single function → 5-agent orchestrated system

Modality: Text-only → Multi-modal (Vision, Voice, Doc AI)

Analytics: None → Text-to-SQL + ARIMA\_PLUS

Legal: None → RAG pipeline + Firestore Vector Search

### Key Refinements

Atomic Bids: Multi-party updates handled in single Firestore batch transaction

3-Tier ML Fallback: ARIMA model → raw aggregation → deterministic pattern

Hybrid AI: Physics-based rules engine kicks in if Gemini times out

Smart Client: Singleton proxy auto-detects API key vs service account

# Gen AI Academy

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# Thank you

## Deployments

### Live Demo

[annapurna-web-887568501843.us-central1.run.app](https://annapurna-web-887568501843.us-central1.run.app)

### Source Code

[github.com/sumitsaraswat362/Annapurna-Gemini-APAC](https://github.com/sumitsaraswat362/Annapurna-Gemini-APAC)

**Save food. Save money. Save the cold chain.**

*Team Annapurna AI — Built for India.  
Powered by Google Cloud.*

## By The Numbers

- 9 Google Cloud Services
- 8 Gemini AI Capabilities
- 5 Autonomous AI Agents
- 14 Server-Side API Routes
- 4 Data Input Modalities
- 14 Days Predictive Forecasting
- <90s Failure to Completed Sale
- 85% Cargo Value Recovered